

Mindsweeper — Competition Submission Report

An agentic AI system for assessing mathematical character, not just mathematical answers. *Precision Learning — not a test, a sweep.*

Project	Mindsweeper — Agentic AI Mathematics Assessment Platform
Competition	Hackastone: Global Contest on AI for Education
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Institution	Nanyang Technological University
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Demo	Video demonstration (embedded in the project website)
Project website	https://mindsweeper.sg
Source code	https://github.com/huidadou/mindsweeper

1. University Introduction

This project originates from **Nanyang Technological University (NTU)**. The work sits at the intersection of cognitive neuroscience, the science of learning, and applied educational technology — translating findings about how the brain learns into practical tools that help learners achieve cognitive breakthroughs.

The team is led by **Zheng Jiahui** (MSc Science of Learning, NIE, NTU, and MSc Technopreneurship and Innovation (Chinese), NTU) and **Gao Siqiao** (MSc Science of Learning, NIE, NTU), under the advisory of **Prof. Cai Yiyu** (NTU), and with technical consultancy from **Wang Quchao** (National University of Singapore, MSc in Data Science and Machine Learning, DSML). The team's research interests span metacognition and intrinsic motivation, cognitive resilience across the lifespan, and the design of assessment that is grounded in validated learning science rather than ad-hoc heuristics. Mindsweeper applies that research base to a concrete, high-value problem in mathematics education.

2. Project Background

2.1 The problem — every learner has hidden mines

Precision Learning starts with an accurate diagnosis — and most assessments find the wrong one. They catch *what* a student got wrong, not *why*, and miss entirely what lies underneath. A mathematics tutor meeting a new student faces a cold-start problem: the student arrives with a grade, perhaps a past paper, and a vague sense of difficulty, and the tutor then spends three to five sessions on informal diagnosis before they have a clear picture. That phase is expensive, inconsistent, and wholly dependent on the individual tutor's skill.

Conventional testing does not solve this. A test produces a **score** — it tells you what failed, never why. Concretely, three things go uncaptured:

1. **Root cause.** A wrong answer tells you the item failed, not the mechanism behind it.
2. **Mindset and motivation.** Fixed mindset, shame-driven effort, and amotivation are all invisible to a mark scheme.
3. **Resilience and metacognition.** Anxiety under pressure and the calibration gap between confidence and performance are never recorded.

The cost of this is concrete: two students who both score 45% on a calculus test can need opposite interventions — one has conceptual gaps and needs reteaching from foundations; the other understands the concepts but freezes under load, drops working steps, and circles the wrong answer despite correct reasoning. A score cannot tell them apart. A human expert can — but only after weeks of observation.

Mindsweeper automates and systematises that expert diagnosis, producing in a single 30-60 minute session what typically takes weeks of human observation.

2.2 What Mindsweeper is

Mindsweeper is an **agentic** AI assessment platform for mathematics learners — *not a test, a sweep*. It spans **K-12, IB, IGCSE, and A Level** (plus adult learners re-engaging with the subject) and runs in a single **30-60 minute** session. It does not deliver a fixed test: it runs as an autonomous assessor that adapts its path in real time through a **perceive → decide → act → reflect** loop, with **eight specialised AI agents working in parallel** around a shared student model.

It triangulates **three independent evidence sources**:

1. **Live chat** — an adaptive Socratic conversation that probes reasoning, not just answers (validated psychological instruments are administered within this phase).
2. **Portfolio** — structured reading of the student's existing work (past papers, classwork).
3. **Transcript** — re-derivation of the full profile from a prior session's transcript.

The output is not a number. It is a structured profile organised across three layers, defined next.

2.3 The assessment framework: Content · Capacity · Control

Mindsweeper profiles **what a learner knows and how they learn**. Every observation belongs to one of three layers, which mirror the cognitive architecture itself — long-term memory, working memory, and executive control:

- **Content (Knowledge & Skill)** — *what the student knows*. Topic-level mastery across all strands, procedural-vs-conceptual split, an error taxonomy (conceptual / procedural / notation / careless), cross-topic transfer, and a readiness indicator. Mapped against the **SAP Coherence Map** of standards.
- **Capacity (Cognition)** — *the working-memory resource the student spends in the moment*. How many steps they can hold, where load overwhelms execution, whether they offload to external memory, how automatic their fluency is. The object level — the cognition doing the mathematics.
- **Control (Self-regulation)** — *how the student governs the above*. This is the **M³ framework — Mindset · Motivation · Metacognition** — plus resilience and affect. The meta level — the cognition watching the cognition.

These layers are produced by the system's specialist agents, whose domains map onto them directly:

Agent domain	Layer	Core signals	Theory anchor
Knowledge & Skill	Content	mastery · errors · transfer	SAP Coherence Map
Mindset & Motivation	Control	SDT spectrum · growth vs. fixed	Dweck (2006); Ryan & Deci (2000)
Resilience & Affect	Control	cliff vs. slope · anxiety	Yeager & Dweck (2012)
Metacognition & Communication	Control	calibration gap · ICAP	Flavell (1979); Chi & Wylie (2014)
Cognition (working memory / load)	Capacity	step capacity · offloading · fluency	Sweller (1988); Baddeley & Hitch

A defining property of the framework is that the layers are *coupled and observable in relation to one another*: a student who writes the givens down before computing (a Control strategy) measurably lowers their Capacity load — a cross-layer relationship that is directly actionable for a teacher and that no single score can surface.

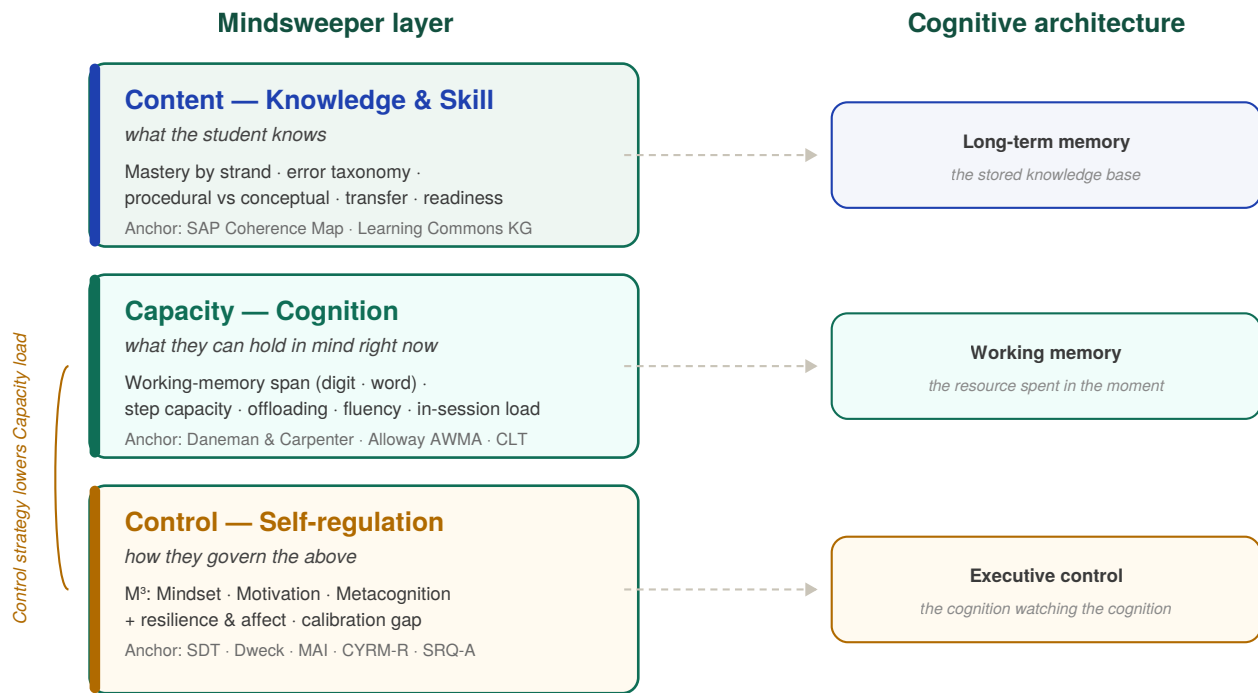


Figure 1 — The Content · Capacity · Control framework, each layer mapped to its cognitive-architecture analogue.

Control — M³ and its instruments. M³ is grounded in self-regulated-learning theory (Zimmerman, 1989; Pintrich, 2004), in which learner agency is exercised through co-regulating metacognitive, motivational, and behavioural components. Each is operationalised with a published, age-matched instrument, and instrument fidelity (exact validated wording, correct age band, correct scoring) is a first-order design constraint:

- **Motivation** — SRQ-A (Self-Regulation Questionnaire — Academic), 16-item short form, scored for the Relative Autonomy Index and four sub-scales (operationalising the Self-Determination Theory spectrum; Ryan & Deci, 2000).
- **Metacognition** — Harrison & Vallin (2018) shortened MAI, 19 items, yielding Knowledge-of-Cognition and Regulation-of-Cognition sub-scales.
- **Resilience / affect** — CYRM-R Personal sub-scale, 10 items.

Capacity — the cognition layer. Working memory and cognitive load are measured with the same triangulated philosophy: a validated self-report anchor (the Paas single-item mental-effort scale, or the Leppink et al. (2013) cognitive-load scale separating intrinsic, extraneous, and germane load), paired with the agent's behavioural inference, and short controlled **working-memory span tasks now administered in every session** — a forward **digit span** (verbal short-term storage) and a **word-span** task (last words recalled across sets growing to five sentences, in the tradition of Daneman & Carpenter's reading span and the Alloway AWMA battery). Measured span and ecological in-session load are kept strictly separate, and their convergence is computed. The agent's behavioural read is itself grounded in the item content of a validated scale (the Working Memory Rating Scale; Alloway, Gathercole & Kirkwood, 2008), so the in-session inference operationalises a published instrument rather than improvising.

On personality traits (OCEAN / Big Five). Personality-level signals — conscientiousness, openness, emotional reactivity — carry genuine diagnostic value for mathematics learning. However, the current prototype contains exploratory Big Five signals that are **not yet governed by the same validated-instrument framework as the M³ constructs**. Responsible integration requires validated instruments, informed consent for minors, and clear governance of how labels are stored and used. The gap between stated position and current build is addressed in §5.2.

2.4 What is novel (*Innovation and Novelty*)

- **It assesses character and cognition, not only correctness.** Any adaptive engine can tell you a student scored 60%. Mindsweeper tells you *why* — whether they panicked, memorised without understanding, are under-confident, or have a genuine gap — which is the information a teacher needs to intervene.
- **It uses a multi-agent blackboard architecture.** Eight specialised roles — each with defined inputs, outputs, and boundaries — are orchestrated on a shared student model. In the current implementation, these roles are realised as structured prompt-level specialisations within the model calls, coordinated by an orchestrator that runs a perceive → decide → act → reflect loop. The assessment methodology and probe taxonomy, not the plumbing, are the core contribution.
- **It triangulates three independent evidence sources** (live chat, portfolio, transcript), with every claim required to cite specific evidence.
- **It performs dynamic assessment, not only static measurement.** In its Choose-your-question mode, the agent *teaches through* a chosen problem with a graduated prompting ladder (Vygotsky's ZPD; Feuerstein), measuring **learning potential** — how far a student moves once supported — and classifying the *questions the student asks* by depth (ICAP; Chi & Wylie, 2014). This captures what a one-shot test structurally cannot: not what the student can do alone, but what they can do next.
- **It re-attributes signals correctly.** Behaviour previously read as "test anxiety" (revisions under pressure, a fatigue threshold, execution collapsing on multi-step work) is, under the Capacity layer, more accurately diagnosed as *working-memory load* — a different and more treatable problem.

2.5 Why it matters (*Educational Significance*)

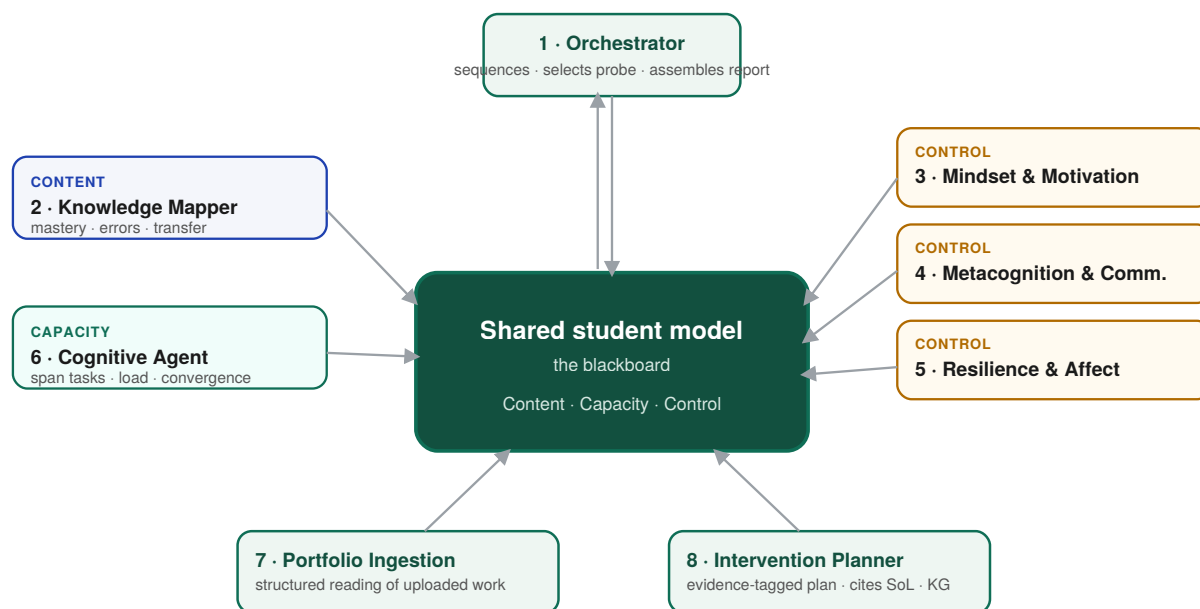
Mindsweeper is built on, and traceable to, the learning-science literature: self-regulated learning (Zimmerman, 1989; Pintrich, 2004); self-determination theory (Ryan & Deci, 2000); mindset and motivation (Dweck, 2006; Yeager & Dweck, 2012); metacognition (Flavell, 1979); cognitive load theory and working memory (Sweller, 1988; Baddeley & Hitch); the ICAP engagement framework (Chi & Wylie, 2014); principles of instruction (Rosenshine, 2012); dynamic assessment and the ZPD (Vygotsky; Feuerstein); and validated instrumentation throughout. Standards and curriculum grounding draws on the SAP Coherence Map, the Learning Commons knowledge graph, LVN, and ANet. For teachers, it compresses weeks of diagnostic observation into one session and returns an evidence-cited, intervention-ready profile. For students, the same data is reframed positively — strengths, growth, and "what to work on next" — in age-appropriate language. The result is assessment that informs teaching, instead of merely ranking learners.

3. Technical Implementation

3.1 Architecture — eight agents, one blackboard

Mindsweeper is a **role-based blackboard architecture**: eight specialised assessment roles — implemented as structured prompt segments within a single model orchestration — read from and write to a **shared student model** (the blackboard), so that evidence from different sources accumulates into one coherent profile. The specialist roles are:

- **Orchestrator** — sequences the assessment and arbitrates which agent acts; assembles the final report, convergence analysis, and verdict.
- **Knowledge Mapper (Knowledge & Skill)** — builds the Content layer (mastery, errors, transfer).
- **Mindset & Motivation** — the SDT-spectrum and growth-vs-fixed reading (Control).
- **Metacognition & Communication** — calibration gap and explanation/ICAP quality (Control).
- **Resilience & Affect** — the cliff-vs-slope and anxiety reading (Control).
- **Cognitive Agent** — the Capacity layer: working-memory span tasks and in-session cognitive-load signals, kept strictly separate and cross-validated.
- **Portfolio Ingestion** — structured reading of uploaded student work.
- **Intervention Planner** — turns the profile into an evidence-tagged plan, citing named science-of-learning principles and consulting the Learning Commons Knowledge Graph for content sequencing.



Specialised agents read from and write to the shared model; the Orchestrator runs a perceive → decide → act → reflect loop, emitting an inspectable trace each turn. Evidence from every source accumulates into one coherent profile.

Figure 2 — The eight-agent blackboard: specialised agents read from and write to one shared student model.

The live **Socratic assessor** runs the perceive → decide → act → reflect loop and a defined **probe taxonomy** (reasoning transparency, perturbation, teaching, metacognitive calibration, frustration, structural reasoning, and a dedicated **Capacity / cognitive-load probe**), emitting a structured `<trace>` on every exchange — the probe deployed, signals detected, an emerging-profile update, and a cognitive-load read. That trace is what makes the agent's reasoning inspectable rather than a black box. The **Cognitive Agent** that produces the Capacity layer (§2.3) administers controlled span tasks (forward digit span; a word-span task) *before* the session and reads ecological load signals *during* it, computing the convergence between the two — a distinction the architecture enforces, since working-memory capacity cannot be read reliably from Socratic dialogue alone.

3.2 Assessment flow

The session phase order is fixed and enforced: **intake → portfolio upload → validated M³ questionnaires (SRQ-A · MAI · CYRM-R) → working-memory span tasks (digit span · word span) → mode selection → live session → report generation**. The three assessment layers never substitute for one another and the phase order is treated as an invariant. Questionnaire items are delivered as an agentic chat-bubble dialogue (sequential bubbles, inline responses), not a static form, so the instruments feel continuous with the conversation; the span tasks are administered as short controlled measures immediately after the self-reports and before mode selection.

3.3 Three deployed surfaces

All three share a common data contract and are served from one origin:

1. **Student assessment agent** — the full session flow, with KaTeX mathematics rendering and a downloadable three-part report (knowledge map, Capacity/M³ profile with banded gauges, convergence chips).
2. **Teacher console** — a longitudinal view: M³ trajectory chart, a Capacity (cognition) panel, knowledge-gap tracking (closed / persisting / new), a learning-potential strip, calibration table, session timeline, and student switcher.
3. **Student console** — the same data reframed for the learner: positive language ("working on next" rather than "gaps", "bouncing back" rather than "resilience"), strength celebration, growth chart, and cross-session encouragement.

3.4 Assessment modes

Two modes, both reading the same Content/Capacity/Control layers:

- **Open Socratic dialogue** — open adaptive probing; *static* assessment (the agent does not teach).
- **Choose-your-question (dynamic assessment)** — the student picks one of three calibrated questions; the agent then teaches *through* it with a graduated prompting ladder, runs a near-transfer item, classifies the student's questions by ICAP depth, and reports learning potential. The level of support required is itself the score. For learners aged 16 and under, the agent adds warmer framing and extra structural prompts within this same mode.

The split is deliberate: Open Socratic is *static* assessment (a clean, uncontaminated baseline); Choose-your-question is *dynamic* assessment (teaching is the measurement, capturing learning potential). An earlier separate "structured" mode was folded into Choose-your-question via age-gated scaffolding, so the two-mode design keeps the static/dynamic distinction crisp without a redundant third path.

3.5 Reliability and reproducibility

Because Mindsweeper is an *assessment* tool, output consistency is a first-order requirement, and large language models are stochastic. The system addresses this architecturally rather than by hoping the model repeats itself.

The core principle is to separate measurement from expression. Reliability in psychometrics requires *construct-level* consistency (is the learner's motivation introjected or autonomous?), not verbatim-identical text. Every component is therefore classed as either deterministic or LLM-mediated:

- **Deterministic by construction** — the validated instruments (SRQ-A Relative Autonomy Index and subscales; MAI Knowledge/Regulation of Cognition; CYRM-R Personal sum) and the span tasks are scored by local code from the learner's own responses. Identical responses yield identical scores. This is the foundation on which reliability rests, and it never touches the model's stochasticity.
- **LLM-mediated, variance-constrained** — the parts the model does drive (Socratic probing, error and character classification, narrative) are controlled three ways: (1) **temperature 0** on every model call, eliminating sampling variance; (2) an **enforced anchor-value schema** — every measurement-bearing field resolves to a fixed enumerated set (e.g. mindset $\in$ {fixed-leaning, mixed, growth-leaning}; resilience pattern $\in$ {cliff, slope, plateau}), so the model's task becomes *selecting the nearest anchor* rather than *describing freely*; unrecognised values are snapped to the nearest validated anchor via explicit synonym matching; values that cannot be resolved are flagged rather than fabricated; and (3) **fixed, evidence-cited rubrics** anchoring each classification to an observed in-session moment.

Converting free description into anchored classification is what makes the measurement outputs *scoreable*: categorical fields can be checked for agreement with Cohen's κ , which free prose cannot. The report schema is additionally expressed as a JSON-Schema tool definition (`emit_report`), ready to be wired as a schema-forced tool call for API-level conformance. In the current build, report extraction uses the schema as a structural guide with client-side coercion; the formal `tool_choice` enforcement is a near-term implementation step.

The adaptive dialogue is deliberately not reproducible, on the same principle as computerised adaptive testing: two administrations legitimately follow different probe paths, while the *inferred profile* is what must be stable. A fixed script would be more reproducible and less valid.

Validation plan. Reliability is validated at the construct level via **test-retest** (same cohort, ~two-week interval, target ≥ 50 learners): **ICC** for the continuous instrument scores and **Cohen's κ** for the categorical classifications, alongside inter-rater agreement

against blinded expert-teacher classifications. A standalone methodology note accompanies this submission.

3.6 Technology stack and environment

- **Frontend:** React / JSX, compiled and validated with **esbuild**; the two console surfaces are static HTML and need no build step.
- **Mathematics rendering:** **KaTeX** (via CDN).
- **Data visualisation:** **Chart.js** (teacher-console trajectories).
- **Model:** the Anthropic API (Claude) drives the agentic reasoning, with **temperature 0** on every call and an enforced anchor-value output schema for measurement fields (see §3.5). The API key is supplied at runtime (stored only in the browser for the demo, or via a server-side proxy in production) and is never committed to source.
- **Data contract:** a browser `localStorage` schema shared across surfaces (`mindsweeper:index`, `mindsweeper:student:<id>`).
- **Hosting:** the surfaces are designed to be served from a single origin (project domain: `mindsweeper.sg`) so the shared `localStorage` contract works across them. The assessment agent and teacher console are deployed and publicly accessible; the student console is designed but **not yet deployed** (it is a near-term deliverable). The working prototype is demonstrated end-to-end in the [accompanying video](#); the public site hosts a project overview and that demonstration.

Setup requirements (also stated in the video): browser-only, no installation for end users; the JSX agent is compiled with esbuild (or run inside the artifact proxy); no API key is bundled; the current `localStorage` contract is same-origin and single-device.

3.7 Source code

GitHub: <https://github.com/huidadou/mindsweeper> . The repository contains the agent, both consoles, the validated-instrument specifications, the shared schema contract, and the multi-agent design documentation.

3.8 Technical Readiness Level (TRL)

Stated honestly: Mindsweeper is at approximately **TRL 4-5** — a working prototype, validated in a relevant environment and demonstrated end-to-end in the accompanying video. The assessment agent and teacher console run end-to-end, and the system has been exercised on real student cases (a worked longitudinal example is built into the teacher console).

The known gap is equally explicit: there is **no backend yet**. The `localStorage` contract does not persist across devices, there is no authentication or per-user identity, and cross-device or multi-user use is not yet supported. Crucially, the hard and novel part — the assessment methodology — is the part that is built; what remains is standard, well-understood engineering (database, auth, hosting), addressed in the roadmap below. This is a deliberate, accurate TRL claim: the concept is proven and the path to production is short and de-risked.

4. Commercial Viability and Economic Impact

4.1 Market size and ceiling

Mindsweeper sits at the intersection of three large, fast-growing markets, which together define its opportunity and its ceiling.

Market	Size (recent)	Forecast	CAGR	Source
Global private tutoring (TAM)	US\$133.8 B (2025)	US\$248.4 B by 2034	7.1%	IMARC (2025)
— corroborating estimate	US\$177.2 B (2024)	US\$295.1 B by 2030	8.9%	Global Industry Analysts (2024)
AI in education (adjacent tailwind)	US\$5.9 B (2024)	US\$32.3 B by 2030	~31%	Grand View Research (2025)
AI tutors (nearest sub-segment)	US\$3.6 B (2025)	US\$6.5 B by 2030	~13%	market est. (2025)

Analyst estimates for private tutoring vary with methodology — from roughly **US\$110-177 B in 2024/25** — but all agree on the shape: a large market growing high-single-digits, with **Asia-Pacific the dominant region (~35.5% share)** and **online/AI-adaptive tutoring the fastest-growing modality (~10-12% CAGR)**. Two structural facts matter for Mindsweeper specifically: **academic training accounts for ~67.8% of application demand** and **the up-to-K-12 segment ~48.7%** — precisely Mindsweeper's target. STEM and test-preparation carry the highest per-student revenue, and these are exactly the high-stakes contexts where accurate diagnosis is most valuable.

TAM → SAM → SOM.

- **TAM** — the global private-tutoring market: **~US\$134 B (2025)**, the total spend Mindsweeper's category ultimately touches.
- **SAM** (*Serviceable Available Market*) — the diagnostic-and-assessment slice of academic K-12 mathematics tutoring, in English-instruction and major-curriculum markets (US Common Core/AP, IB, IGCSE/A-Level, Singapore, and comparable systems). Diagnosis is not separately invoiced today; it is bundled into the first 3-5 tutoring sessions. Taking academic tutoring (~68% of TAM) and attributing a conservative **10-15%** of early-relationship spend to the diagnostic phase yields a serviceable diagnostic slice on the order of **US\$9-14 B**.
- **SOM** (*Serviceable Obtainable Market*) — a realistic near-term beachhead: Singapore and regional international-curriculum schools and tutoring centres, plus a direct-to-consumer web offering. Singapore's private-tutoring spend alone is estimated in the low single-digit billions SGD; capturing even a fraction of the diagnostic layer across an initial cluster of institutions is a multi-million-dollar obtainable market — enough to validate unit economics before scaling into the broader SAM.

4.2 Where the cost is, and how Mindsweeper reduces it — per stakeholder

The economic thesis is simple: **diagnosis is the expensive, unbilled, inconsistent bottleneck of personalised learning, and Mindsweeper automates it.** The value is quantifiable for each stakeholder.

Tutoring institutions / centres. A tutor meeting a new student spends **3-5 sessions** on informal diagnosis before teaching effectively — time that is either unbilled or billed as low-value. At a modest blended tutor cost of **US\$30-60/hour**, that is **US\$90-300 of diagnostic labour per new student**, repeated for every enrolment, and dependent on the individual tutor's skill. Mindsweeper compresses this into **one automated ~30-minute session** whose marginal cost is a few cents of model inference. *Back-of-envelope:* a centre onboarding 500 new students a year at 4 diagnostic sessions × US\$45/session ≈ **US\$90,000/year in diagnostic labour** — a cost Mindsweeper largely removes while making the output more consistent and evidence-cited. The saved tutor hours convert directly into billable teaching capacity.

Teachers (schools). Teacher time is the scarcest resource in education. Teachers using AI tools weekly report saving **~5.9 hours per week** on average (equivalent to roughly six working weeks per school year). Mindsweeper targets one of the most time-consuming judgement tasks — building an accurate picture of *why* a struggling student is struggling — and returns an evidence-cited profile that would otherwise take weeks of observation. For a teacher with 150 students, even a partial automation of diagnostic and progress-monitoring effort is a material reclamation of instructional time.

Parents. Parents currently pay for diagnostic sessions indirectly (bundled into tutoring fees) and often pay *twice* — once for a misdiagnosis, again for the corrective tutoring. Two students scoring 45% can need opposite interventions; a parent who buys "more practice" when the real problem is working-memory overload or exam anxiety has wasted the spend. Mindsweeper reduces this **misallocation cost**: an accurate up-front profile means tutoring money is spent on the right intervention from the start.

Students. The cost to students is time and morale spent on mis-targeted work. Accurate diagnosis shortens the path to improvement and — through the Control layer — protects motivation and resilience, the factors most predictive of whether a learner persists at all.

Mindsweeper's own unit cost. The other side of the equation is what a Mindsweeper assessment *costs to run*. The live agent uses Claude Sonnet (Anthropic API), billed per token at US\$3 per million input tokens and US\$15 per million output tokens, with prompt caching reducing repeated input (such as the system prompt) by up to 90%. A full ~30-minute adaptive session is on the order of fifteen conversational turns plus a portfolio-analysis pass. On our measured prompt sizes and turn counts, this works out to approximately **US\$0.20-0.50 per completed assessment** with prompt caching enabled; routing the mechanical sub-tasks (portfolio and transcript analysis, report assembly) to the lower-cost Claude Haiku tier (US\$1/US\$5 per million tokens) while keeping the Socratic reasoning on Sonnet brings the typical case toward **~US\$0.20-0.30**. (These are back-of-envelope figures from our own token accounting at list prices; the Batch API's 50% discount does not apply to live interactive sessions but is available for any offline reprocessing.)

The comparison is the whole argument: an assessment that costs **cents** replaces a manual diagnostic process that costs **tens to hundreds of dollars** in tutor labour — a marginal cost roughly three orders of magnitude below the alternative it displaces. This is the structural basis of the margin claimed in §4.3.

4.3 Entrepreneurial framing — the viability triangle

A venture is viable only where three conditions hold simultaneously — the classic **desirability × feasibility × viability** intersection:

- **Desirability (is it wanted?)** — Strong. The diagnostic pain is universal and currently unsolved; personalised and adaptive learning is the number-one demand driver cited across every market report above; and no incumbent assessment tool explains *why* rather than *what*. Character-and-capacity diagnosis is a felt need for teachers, tutors, parents, and students alike.
- **Feasibility (can we build it?)** — Demonstrated. The hard part — the assessment methodology — is built and running at **TRL 4-5**; what remains is standard engineering (§3.8). The reliability architecture (§3.5) makes the output defensible as a measurement, not a novelty.
- **Viability (can it sustain a business?)** — Favourable. A near-zero marginal cost per assessment (model inference) against a task that currently costs US\$90–300 in human labour gives a wide margin. Standard SaaS monetisation applies.

Where all three overlap is the venture. Mindsweeper's distinctive position is that its **desirability** rests on a capability (explaining *why*) that competitors structurally lack, and its **viability** rests on collapsing a high-cost manual process to marginal-cost automation.

A useful way to express the opportunity:

Market viability $\approx$ **market ceiling** $\times$ **obtainable share** $\times$ **margin** $\times$ **defensibility**.

- **Market ceiling** — a ~US\$134 B tutoring TAM growing ~7–9%, with an AI-in-education tailwind compounding at ~31%.
- **Obtainable share** — begins with a focused SOM (Singapore + international-curriculum beachhead) and expands into the ~US\$9–14 B diagnostic SAM.
- **Margin** — high: assessment marginal cost is roughly US\$0.20–0.50 (§4.2); the displaced alternative costs tens-to-hundreds of dollars in tutor time — a ~1,000× gap.
- **Defensibility** — the moat is the methodology (validated-instrument grounding, the agentic probe taxonomy, the Content/Capacity/Control framework, the reliability layer, and dynamic assessment), not the replicable application plumbing.

4.4 Productisation and monetisation path

The architecture is designed to be exposed as a service, and the same underlying engine supports two commercial motions in parallel:

- **B2B (institutions and organisations)**. Mindsweeper can be **customised for schools, tutoring centres, and education organisations** — mapped to their curriculum, branded, and integrated into their existing workflow via a planned **MCP**

(Model Context Protocol) server that lets the assessment agents be invoked as tools inside other platforms. This supports institutional licences and partnership/embedding models — the technology-licensing pattern now visible across the industry (e.g. platform providers licensing hybrid-delivery AI to expand internationally without capital-intensive market entry).

- **B2C (tutors, parents, and students).** The same engine powers a **direct web-app offering (mindsweeper.sg)** for individual tutors, parents, and learners (including adults re-engaging with the subject), monetised on a per-seat / subscription basis.

Monetisation therefore follows standard SaaS tiers: **institutional licences for schools and centres (B2B), and per-seat / direct-to-consumer subscriptions for tutors, parents, and students (B2C).** Because the diagnostic step is currently unpriced — bundled invisibly into tutoring fees — Mindsweeper is not competing on price against an incumbent line item; it converts an invisible cost into a visible, lower-cost product, which is an easier sale than displacing an existing paid tool.

5. Known Limitations (v0.5)

#	Problem	Why	Proposed remedy
1	Multi-agent architecture. The eight assessment roles are prompt-level specialisations within the model calls, not eight independently deployed services.	Prototype priority was proving the assessment methodology (probe taxonomy, boundary rules, convergence logic), not production orchestration infrastructure.	Describe as "role-based orchestration on a shared student model" (done in §2.4). Multi-service deployment is a production goal.
2	Personality signals (OCEAN). Exploratory Big Five signals (including neuroticism) remain in the codebase — report schema, prompt mappings, and teacher-console character view — despite the report's stated exclusion of personality inference.	Personality-level signals (emotional reactivity under cognitive load) have real diagnostic value, but inferring a Big Five profile for a named minor from one session is psychometrically weak and ethically serious. The current build kept the capability without the safeguards: <code>ocean_signals</code> is the least-governed field in the system.	<i>Immediately:</i> remove <code>ocean_signals</code> from all output and persisted records. <i>Going forward:</i> reintroduce only with a validated instrument, the same anchored-enum governance as M ³ , explicit informed consent for minors, and never surfaced as a trait label to the student.
3	Report schema enforcement. The <code>emit_report</code> JSON-Schema tool definition exists but is not wired as a forced <code>tool_choice</code> call. Report extraction uses a structural guide with client-side coercion, not API-guaranteed conformance.	The schema was authored and staged; wiring it into the request body was not completed in time. The coercion layer snaps unrecognised values to the nearest anchor rather than returning <code>null</code> .	Wire <code>emit_report</code> as a forced tool call so conformance is API-enforced. Change coercion to return <code>null</code> on unrecognised values. Until then, described as "schema-guided with client-side validation."
4	Student console. The student-facing console (age-appropriate, growth-oriented language) is designed but not yet deployed.	Development prioritised the assessment agent and teacher console; the student console shares the same data contract and was sequenced last.	Deploy shortly on the same origin. Until live, the student-view entry point is hidden rather than left as a broken link.
5	Live agent access. The live agent requires a user-supplied Anthropic API key held in the browser. In practice, browsers block the direct call entirely (CORS restriction Anthropic enforces to prevent key exposure in web pages), so the deployed live build cannot complete a real session.	A static GitHub Pages site has no backend to hold a server-side key or proxy the call. Direct-from-browser calling is the only option without infrastructure — and Anthropic blocks exactly that. Target users (students, parents, teachers) do not hold API keys.	Move the model call to a server-side proxy (§6, roadmap item 3). The key stays on the server; the client authenticates via a short-lived session token. Until then, the demo build (no key required) is the recommended entry point.

#	Problem	Why	Proposed remedy
6	Working-memory span tasks. Single trial per set-size for both digit span and word span; digit-span ladder pushes to lengths 9–10 without an adaptive stop.	The span-task module was the last component built; a minimal single-trial version was the fastest route to a measured signal in the session flow.	2–3 trials per set-size, partial-credit-unit scoring (Conway et al., 2005), standard termination rule, age-appropriate starting length, no adjacent-repeat digits. Priority for the validation study.
7	Mobile responsiveness. Teacher console overflows on narrow viewports (~390px).	Console was designed desktop-first; responsive pass was deprioritised against the core diagnostic flow.	Mobile layout for top bar and student selector (vertical stacking, single-column cards). Assessment agent and landing page are already mobile-functional.

6. Roadmap

Immediate next milestone, converting the validated prototype into a multi-user product:

1. **Persistence** — replace `localStorage` with a real database (e.g., Supabase / Postgres), giving longitudinal per-student records.
2. **Identity** — add authentication and personal IDs (the per-user accounts the product needs).
3. **Secure model access** — move the Anthropic API call server-side via a serverless function, so the key lives on the server and browser-side CORS restrictions no longer apply (see §5.5).
4. **Hosting** — deploy frontend and backend together on mindsweeper.sg, including the student console (§5.4).

This milestone is a matter of weeks, not months, and requires no change to the assessment methodology — only the infrastructure around it.

Validation research (parallel track). Beyond engineering, the priority research milestone is a formal **validation and reliability study** (as set out in §3.5): a **test-retest** study with a target cohort of **≥ 50 learners (Year 7+)** across curricula, reporting **ICC** for the continuous instrument scores and **Cohen's κ** for the categorical classifications, alongside inter-rater agreement against blinded expert-teacher judgements (construct validity) and internal-consistency checks (Cronbach's α) on the shortened instrument subsets. This study is what converts Mindsweeper from a demonstrably-working prototype into an **evidence-backed, publishable measurement instrument** — the credibility layer that matters most to institutional (B2B) buyers.

Subsequent engineering work: hardening the report schema enforcement (§5.3), expanding the working-memory span tasks (§5.6), the personality-assessment redesign (§5.2), and the

MCP service layer. The Capacity-layer span-task module (digit span and word span) is implemented and integrated into the session flow.

Appendix — Key references

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